10.4 (a) By replacing ε_r with $\varepsilon_r + i\sigma/\omega\varepsilon_0 = \varepsilon_r + iZ_0\sigma/k$, and using Eqs. (10.10) and (10.11), the differential and total scattering cross sections are, respectively.

$$\frac{d\sigma_{sc}}{d\Omega} = \frac{1}{2}k^4a^6(1+\cos^2\theta)\left|\frac{\varepsilon_r-1}{\varepsilon_r+2}\right|^2 = \frac{1}{2}k^4a^6(1+\cos^2\theta)\,\frac{(\varepsilon_r-1)^2+(Z_0\sigma/k)^2}{(\varepsilon_r+2)^2+(Z_0\sigma/k)^2},$$

$$\sigma_{sc} = \frac{8\pi}{3}k^4a^6\,\frac{(\varepsilon_r-1)^2+(Z_0\sigma/k)^2}{(\varepsilon_r+2)^2+(Z_0\sigma/k)^2}.$$

(b) From Eq. (10.134), the absorbed power is

$$P_{abs} = -\frac{1}{2\mu_0}\oint_{S_1}\mathrm{Re}[\vec{E}\times\vec{B}^*]\cdot\vec{n}'\,da' = -\frac{1}{2}\mathrm{Re}\int\nabla\cdot(\vec{E}\times\vec{H}^*)\,d^3x$$

$$= -\frac{1}{2}\mathrm{Re}\int\left[\vec{H}^*\cdot(\nabla\times\vec{E}) - \vec{E}\cdot(\nabla\times\vec{H}^*)\right]d^3x$$

$$= -\frac{1}{2}\mathrm{Re}\int\left[\vec{H}^*\cdot\left(-\frac{\partial\vec{B}}{\partial t}\right) - \vec{E}\cdot\vec{J}^* - \vec{E}\cdot\frac{\partial\vec{D}^*}{\partial t}\right]d^3x$$

$$= \frac{1}{2}\mathrm{Re}\int\vec{J}^*\cdot\vec{E}\,d^3x + \frac{1}{2}\mathrm{Re}\int\left(\vec{H}^*\cdot\frac{\partial\vec{B}}{\partial t} + \vec{E}\cdot\frac{\partial\vec{D}^*}{\partial t}\right)d^3x$$

$$= \frac{1}{2}\int\sigma|\vec{E}|^2d^3x + \frac{1}{4}\frac{\partial}{\partial t}\int\left(\frac{1}{\mu_0}|\vec{B}|^2 + \varepsilon_0|\vec{E}|^2\right)d^3x$$

The second term is zero, as the total energy is constant in the sphere. From Eq. (4.53), the electric field inside the sphere is constant and is given by

$$\vec{E} = \frac{3}{\varepsilon_r+2}\vec{E}_i.$$

Therefore. $$P_{abs} = \frac{\sigma}{2}\cdot\frac{9}{|\varepsilon_r+2|^2}|\vec{E}_i|^2\cdot\frac{4}{3}\pi R^3 = \frac{6\sigma\pi R^3}{(\varepsilon_r+2)^2+(Z_0\sigma/k)^2}|\vec{E}_i|^2$$

and $$\sigma_{abs} = \frac{P_{abs}}{P_i} = \frac{6\sigma\pi R^3}{(\varepsilon_r+2)^2+(Z_0\sigma/k)^2}|\vec{E}_i|^2\Big/\frac{|\vec{E}_i|^2}{2Z_0}$$

$$= 12\pi R^2\,\frac{RZ_0\sigma}{(\varepsilon_r+2)^2+(Z_0\sigma/k)^2}$$

(c) The scattered electric field is

$$\vec{E}_{sc} = \frac{1}{4\pi\varepsilon_0}k^2\frac{e^{ikr}}{r}\left((\vec{n}\times\vec{p})\times\vec{n}\right),\quad \text{where } \vec{p} = 4\pi\varepsilon_0\frac{\varepsilon_r-1}{\varepsilon_r+2}R^3\vec{E}_i.$$

In terms of the incident field,

$$\vec{E}_{sc} = \frac{e^{ikr}}{r} E_0 k^2 R^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \left((\vec{n} \times \vec{\varepsilon}_0) \times \vec{n} \right).$$

By definition, $\vec{f}(\vec{k}, \vec{k}_0) = k^2 R^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \left((\vec{n} \times \vec{\varepsilon}_0) \times \vec{n} \right).$

Using the optical theorem,

$$\sigma_t = \frac{4\pi}{k} \operatorname{Im}\left[\vec{\varepsilon}_0^{\,*} \cdot \vec{f}(\vec{k} = \vec{k}_0) \right] = \frac{4\pi}{k} \operatorname{Im}\left[k^2 R^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \vec{\varepsilon}_0^{\,*} \cdot \left((\vec{n} \times \vec{\varepsilon}_0) \times \vec{n} \right) \Big|_{\vec{n} = \vec{n}_0} \right]$$

$$= 4\pi k R^3 \operatorname{Im}\left[\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \left| \vec{n}_0 \times \vec{\varepsilon}_0 \right|^2 \right] = 4\pi k R^3 \operatorname{Im}\left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \right)$$

$$= 4\pi k R^3 \frac{3 Z_0 \sigma / k}{(\varepsilon_r + 2)^2 + (Z_0 \sigma / k)^2} = 12\pi R^2 \frac{R Z_0 \sigma}{(\varepsilon_r + 2)^2 + (Z_0 \sigma / k)^2}.$$

This only agrees with the absorption cross section, as scattering cross section is of higher order in long-wavelength limit.